

**Correct Marks : 4**

Question Label : Multiple Choice Question

A financial advisor tells you that you can become a millionaire if you start investing early. You start to deposit 25,000 units every year in a savings account which earns 8% annual interest. What is the earliest year at which your account balance reaches a million units? (Assume you don't make any withdrawals in between)

**Options :**

6406533360464. ✖ 40

6406533360465. ✖ 25

6406533360466. ✔ 19

6406533360467. ✖ 15

## Game Theory

|                                                              |              |
|--------------------------------------------------------------|--------------|
| Section Id :                                                 | 64065369398  |
| Section Number :                                             | 14           |
| Section type :                                               | Online       |
| Mandatory or Optional :                                      | Mandatory    |
| Number of Questions :                                        | 6            |
| Number of Questions to be attempted :                        | 6            |
| Section Marks :                                              | 25           |
| Display Number Panel :                                       | Yes          |
| Section Negative Marks :                                     | 0            |
| Group All Questions :                                        | No           |
| Enable Mark as Answered Mark for Review and Clear Response : | No           |
| Section Maximum Duration :                                   | 0            |
| Section Minimum Duration :                                   | 0            |
| Section Time In :                                            | Minutes      |
| Maximum Instruction Time :                                   | 0            |
| Sub-Section Number :                                         | 1            |
| Sub-Section Id :                                             | 640653146759 |
| Question Shuffling Allowed :                                 | No           |

**Question Number : 290 Question Id : 640653995377 Question Type : MCQ**

**Correct Marks : 0**

Question Label : Multiple Choice Question

**THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : GAME THEORY (COMPUTER BASED EXAM)"**

**ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?**

**CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.**

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE [TOP](#) FOR THE SUBJECTS REGISTERED BY YOU)

**Options :**

6406533360468. ✓ YES

6406533360469. ✗ NO

**Sub-Section Number :**

2

**Sub-Section Id :**

640653146760

**Question Shuffling Allowed :**

No

**Question Id : 640653995378 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (291 to 292)**

Question Label : Comprehension

An item is up for auction. Player 1 values the item at 3 while player 2 values the item at 5. Each player can bid either 0, 1 or 2. If player  $i$  bids more than player  $j$  then  $i$  wins the good and pays his bid, while the loser does not pay. If both players bid the same amount, then a coin is tossed to determine who the winner is, who gets the good and pays his bid while the loser pays nothing. Formulate the game in matrix form and answer the given subquestions.

**Sub questions**

**Question Number : 291 Question Id : 640653995379 Question Type : MCQ**

**Correct Marks : 1**

Question Label : Multiple Choice Question

Which player does not have strictly dominated strategy?

**Options :**

6406533360470. ✓ Player 1

6406533360471. ✗ Player 2

6406533360472. ✗ None

**Question Number : 292 Question Id : 640653995380 Question Type : MCQ**

**Correct Marks : 2**

Question Label : Multiple Choice Question

Which strategies survive Iterated Elimination of Strictly Dominated Strategy?

**Options :**

6406533360473. ✗ (0, 0)

6406533360474. ✗ (1, 1)

6406533360475. ✓ (2, 2)

6406533360476. ✗ None

**Sub-Section Number :**

3

**Sub-Section Id :**

640653146761

**Question Shuffling Allowed :**

No

**Question Id : 640653995384 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (293 to 298)**

Question Label : Comprehension

Determined to crack down on the drug trade, a city mayor puts more officers out on patrol to disrupt the business of drug dealers. A drug dealer in a neighborhood can work his trade either on a street corner or in the park. Each day, he decides where to set up shop, knowing that word about his location will travel among users. Because a good snitch is lacking, word does not travel to the police. The police officer on the beat then needs to decide whether she will patrol the park or the street corner, while not knowing where the drug dealer is hanging out that day. The decision of the officer and the dealer determine the extent of drug trades that day. Let suppose the total number of potential trades in the market is 100. A dealer's payoff is determined by the number of trades he consummates, while the officer's payoff is based on the number of trades she disrupts. The table below illustrates the payoffs for both parties according to their respective strategies in the game.

|                |               | Drug Dealer   |        |
|----------------|---------------|---------------|--------|
| Police Officer |               | Street Corner | Park   |
|                | Street Corner | 80, 20        | 0, 100 |
|                | Park          | 10, 90        | 60, 40 |

Based on the above data, answer the given subquestions.

### Sub questions

**Question Number : 293 Question Id : 640653995385 Question Type : MCQ**

**Correct Marks : 1**

Question Label : Multiple Choice Question

Choose the correct statement

**Options :**

6406533360485. ✖ There is a unique Nash equilibrium in pure strategy

6406533360486. ✖ There are finitely many Nash equilibria in this game in pure strategy

6406533360487. ✔ There is no Nash equilibrium in this game in pure strategy

6406533360488. ✖ None

**Question Number : 294 Question Id : 640653995386 Question Type : MCQ**

**Correct Marks : 1**

Question Label : Multiple Choice Question

Choose the correct statement

**Options :**

6406533360489. ✔ There is a unique Nash equilibrium in this game

6406533360490. ✖ There are finitely many Nash equilibria in this game

6406533360491. ✖ There is no Nash equilibrium in this game

6406533360492. ✖ None

**Question Number : 295 Question Id : 640653995387 Question Type : SA**

**Correct Marks : 1**

Question Label : Short Answer Question

If the randomization probability of a police officer for patrolling the streets is  $p/13$ , then  $p =$  \_\_\_\_\_

**Response Type : Numeric**

**Evaluation Required For SA : Yes**

**Show Word Count : Yes**

**Answers Type : Equal**

**Text Areas : PlainText**

**Possible Answers :**

5

**Question Number : 296 Question Id : 640653995388 Question Type : SA**

**Correct Marks : 1**

Question Label : Short Answer Question

If the randomization probability of a police officer for patrolling the park is  $q/13$ , then  $q =$  \_\_\_\_\_

**Response Type : Numeric**

**Evaluation Required For SA : Yes**

**Show Word Count : Yes**

**Answers Type : Equal**

**Text Areas : PlainText**

**Possible Answers :**

8

**Question Number : 297 Question Id : 640653995389 Question Type : SA**

**Correct Marks : 1**

Question Label : Short Answer Question

If the randomization probability of the drug dealer choosing street is  $r/13$ , then  $r =$  \_\_\_\_\_

**Response Type : Numeric**

**Evaluation Required For SA : Yes**

**Show Word Count : Yes**

**Answers Type : Equal**

**Text Areas : PlainText**

**Possible Answers :**

6

**Question Number : 298 Question Id : 640653995390 Question Type : SA**

**Correct Marks : 1**

Question Label : Short Answer Question

If the randomization probability of the drug dealer choosing park is  $s/13$ , then  $s =$  \_\_\_\_\_

**Response Type : Numeric**

**Evaluation Required For SA : Yes**

**Show Word Count : Yes**

**Answers Type : Equal**

**Text Areas : PlainText**

**Possible Answers :**

7

**Sub-Section Number :**

4

**Sub-Section Id :**

640653146762

**Question Shuffling Allowed :**

No

**Question Id : 640653995391 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (299 to 306)**

Question Label : Comprehension

Consider a two-player game in which player 1 can choose A or B. The game ends if he chooses A while it continues to player 2 if he chooses B. Player 2 can either choose C or D, with the game ending after C and continuing again with player 1 after D. Player 1 then can choose E or F, and then the game ends after each of these choices.

Imagine that the payoffs following choice A by player 1 are (2, 0), following C by player 2 are (3, 1), following E by player 1 are (0, 0) and following F by player 1 are (1, 2)

Model this as an extensive form game and answer the given subquestions:

**Sub questions**

**Question Number : 299 Question Id : 640653995392 Question Type : MCQ**

**Correct Marks : 1**

Question Label : Multiple Choice Question

This is a game of

**Options :**

6406533360497. ✓ Perfect information

6406533360498. ✗ Imperfect information

6406533360499. ✗ None

6406533360500. ✗ Can't be determined

**Question Number : 300 Question Id : 640653995393 Question Type : MCQ**

**Correct Marks : 1**

Question Label : Multiple Choice Question

How many terminal nodes does this game have?

**Options :**

6406533360501. ✖ 3

6406533360502. ✖ 2

6406533360503. ✔ 4

6406533360504. ✖ 5

**Question Number : 301 Question Id : 640653995394 Question Type : MCQ**

**Correct Marks : 1**

Question Label : Multiple Choice Question

How many information sets does this game have?

**Options :**

6406533360505. ✖ 2

6406533360506. ✔ 3

6406533360507. ✖ 4

6406533360508. ✖ 1

**Question Number : 302 Question Id : 640653995395 Question Type : MCQ**

**Correct Marks : 1**

Question Label : Multiple Choice Question

How many pure strategies does player 1 have?

**Options :**

6406533360509. ✖ 1

6406533360510. ✖ 2

6406533360511. ✖ 3

6406533360512. ✔ 4

**Question Number : 303 Question Id : 640653995396 Question Type : MCQ**

**Correct Marks : 1**

Question Label : Multiple Choice Question

How many pure strategies does player 2 have?

**Options :**

6406533360513. ✖ 1

6406533360514. ✔ 2

6406533360515. ✖ 3

6406533360516. ✖ 4

**Question Number : 304 Question Id : 640653995397 Question Type : MCQ**

**Correct Marks : 1**

Question Label : Multiple Choice Question

How many pure strategy Nash equilibria does this game have?

**Options :**

6406533360517. ✖ 1

6406533360518. ✖ 2

6406533360519. ✔ 3

6406533360520. ✖ 4

**Question Number : 305 Question Id : 640653995398 Question Type : MCQ**

**Correct Marks : 1**

Question Label : Multiple Choice Question

Which one of the Nash equilibria is also a subgame perfect equilibrium?

**Options :**

6406533360521. ✖ (AE, D)

6406533360522. ✔ (AF, D)

6406533360523. ✖ (BE, C)

6406533360524. ✖ (BF, C)

**Question Number : 306 Question Id : 640653995399 Question Type : MCQ**

**Correct Marks : 1**

Question Label : Multiple Choice Question

No. of sub games this game has

**Options :**

6406533360525. ✖ 1

6406533360526. ✖ 2

6406533360527. ✔ 3

6406533360528. ✖ 4

**Sub-Section Number :**

5

**Sub-Section Id :**

640653146763

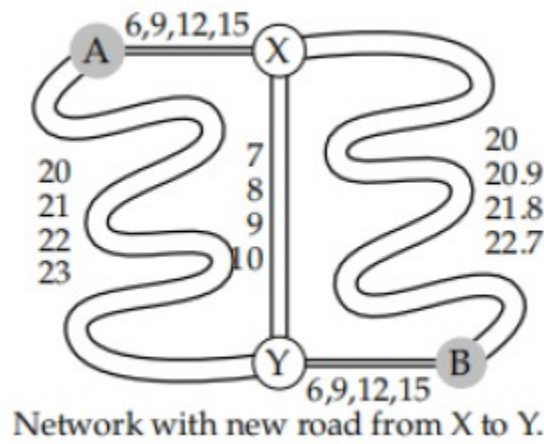
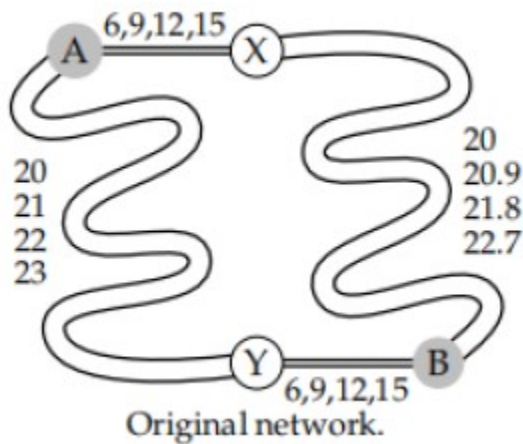
**Question Shuffling Allowed :**

No

**Question Id : 640653995381 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (307 to 308)**

Question Label : Comprehension

**(Choosing a route)** Four people must drive from A to B at the same time. Two routes are available, one via X and one via Y (Refer to the left panel of following Figure). The roads from A to X, and from Y to B are both short and narrow; in each case, one car takes 6 minutes, and each additional car increases the travel time per car by 3 minutes. (If two cars drive from A to X, for example, each car takes 9 minutes.) The roads from A to Y, and from X to B are long and wide; on A to Y one car takes 20 minutes, and each additional car increases the travel time per car by 1 minute; on X to B one car takes 20 minutes, and each additional car increases the travel time per car by 0.9 minutes. Formulate this situation as a strategic game and answer the given subquestions.



## Sub questions

**Question Number : 307 Question Id : 640653995382 Question Type : MCQ**

**Correct Marks : 2**

Question Label : Multiple Choice Question

Choose the correct alternative

**Options :**

6406533360477. ✖ In Nash equilibrium two people take each route

6406533360478. ✖ For the NE action profile, each person's travel time is either 29.9 or 30 minutes

6406533360479. ✔ In Nash equilibrium two people take each route and For the NE action profile, each person's travel time is either 29.9 or 30 minutes

6406533360480. ✖ None

**Question Number : 308 Question Id : 640653995383 Question Type : MCQ**

**Correct Marks : 2**

Question Label : Multiple Choice Question

Now suppose that a relatively short, wide road is built from X to Y, giving each person four options for travel from A to B: A-X-B, A-Y-B, A-X-Y-B, and A-Y-X-B (As shown in the right panel of the given figure). Assume that a person who takes A-X-Y-B travels the A-X portion at the same time as someone who takes A-X-B, and the Y-B portion at the same time as someone who takes A-Y-B. (Think of there being constant flows of traffic.) On the road between X and Y, one car takes 7 minutes and each additional car increases the travel time per car by 1 minute. Find the Nash equilibrium in this new situation and answer the questions.

Choose the correct alternative

**Options :**

6406533360481. ✔ In any Nash equilibrium, one person takes A-X-B, two people take A-X-Y-B, and one person takes A-Y-B

6406533360482. ✖ In this equilibrium profile, each person's travel time is 36 minutes

6406533360483. ✖ In any Nash equilibrium, one person takes A-X-B, two people take A-X-Y-B, and one person takes A-Y-B and In this equilibrium profile, each person's travel time is 36

minutes

6406533360484. ✖ None

**Question Id : 640653995400 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix**

**Question Numbers : (309 to 310)**

Question Label : Comprehension

*(Synergies)* Two division managers can invest time and effort in creating a better working relationship. Each invests  $e_i \geq 0$ , and if both invest more then both are better off, but it is costly for each manager to invest. In particular, the payoff function for player  $i$  from effort levels  $(e_i, e_j)$  is  $v_i(e_i, e_j) = (a + e_j)e_i - e_i^2$ .

Based on the above data, answer the given subquestions.

**Sub questions**

**Question Number : 309 Question Id : 640653995401 Question Type : MCQ**

**Correct Marks : 2**

Question Label : Multiple Choice Question

What is the best response function of player  $i$  for all  $e_i \geq 0$

**Options :**

6406533360529. ✖  $BR_i(e_j) = \frac{(a+e_j)^2}{2}$

6406533360530. ✖  $BR_i(e_i) = \frac{(a+e_i)^2}{2}$

6406533360531. ✔  $BR_i(e_j) = \frac{(a+e_j)}{2}$

6406533360532. ✖  $BR_i(e_i) = \frac{(a+e_i)}{2}$

**Question Number : 310 Question Id : 640653995402 Question Type : MCQ**

**Correct Marks : 2**

Question Label : Multiple Choice Question

Choose the correct alternative

**Options :**

Under Nash equilibrium both players choose

6406533360533. ✖  $e_1 = e_2 = a$

6406533360534. ✖ This is a unique Nash equilibrium